

A CERTIFIED $n = 7$ CASE OF A GENERALIZED STACK-SORTING ENUMERATION CONJECTURE

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ABSTRACT. For the generalized stack-sorting map $s_{123,132}$, Zhang conjectured that the numbers of minimally-sorted permutations satisfy the relation $|M_{2n}| = (n+1)|M_{2n-1}|$. Zhang reported computational verification for $n \leq 6$. We certify the next case. An exact quotient by the labels of the large entries reduces lengths 13 and 14 to 1,235,520 and 2,162,160 skeletons, respectively. Complete enumeration gives $|M_{13}| = 47,265,120$ and $|M_{14}| = 378,120,960$. Moreover, deleting the last entry and standardizing is an eight-to-one map from M_{14} to M_{13} . Two discovery enumerators and a no-import verifier independently reconstruct the integer certificate. This establishes only the finite case $n = 7$, not the full conjecture.

1. INTRODUCTION

Berlow defined and studied the T -restricted stack map s_T and proved that the periodic points of $s_{123,132}$ are exactly the half-decreasing permutations [1, Theorem 4.5]. Zhang subsequently proved

$$\text{ord}_{s_{123,132}}(S_N) = 2 \left\lfloor \frac{N-1}{2} \right\rfloor$$

and defined the minimally-sorted permutations M_N to be those whose transient to periodicity attains this maximum [2, Theorem 1.2]. Zhang's Conjecture 4.4 states

$$(1) \quad |M_{2n}| = (n+1)|M_{2n-1}| \quad (n \geq 1),$$

and the paper reports a computational check for $n \leq 6$ [2, Conjecture 4.4 and the preceding sentence].

Our result is the first relation beyond that reported range.

Theorem 1.1. *For the generalized stack-sorting map $s_{123,132}$,*

$$|M_{13}| = 47,265,120, \quad |M_{14}| = 378,120,960.$$

Consequently, $|M_{14}| = 8|M_{13}|$.

The enumeration also yields a natural finite bijective explanation. If $D(\pi)$ denotes the standardization of the word obtained by deleting the last entry of $\pi \in M_{14}$, then $D : M_{14} \rightarrow M_{13}$ is eight-to-one. The eight inverse choices are terminal insertions of the values $7, 8, \dots, 14$.

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2. DEFINITIONS

A stack is read from top to bottom. The right-greedy map $s_{123,132}$ pushes the next input entry when the resulting stack avoids both classical patterns 123 and 132; otherwise it pops the current stack top to the output and tries the same input entry again. It flushes the stack when the input is empty.

A permutation $\rho \in S_N$ is periodic when $s_{123,132}^r(\rho) = \rho$ for some $r \geq 1$. Put

$$t_N(\pi) = \min\{k \geq 0 : s_{123,132}^k(\pi) \text{ is periodic}\}$$

and

$$M_N = \left\{ \pi \in S_N : t_N(\pi) = 2 \left\lfloor \frac{N-1}{2} \right\rfloor \right\}.$$

Writing $m = \lfloor (N-1)/2 \rfloor$, a permutation is *half-decreasing* when

$$\pi_{N-1}\pi_{N-3}\cdots\pi_{3-(N \bmod 2)} = 12\cdots m.$$

Berlow's theorem identifies this condition with periodicity [1, Theorem 4.5].

Lemma 2.1. *Pushing an input entry x is legal if and only if at most one current stack entry exceeds x .*

Proof. The current stack already avoids both forbidden patterns. A newly created forbidden occurrence must therefore use x as its first, topmost entry. It is a 123 or 132 occurrence precisely when x is smaller than two later stack entries; the relative order of those two entries distinguishes the two patterns. $\square$

3. THE LARGE-LABEL QUOTIENT

Fix a threshold m . For a permutation π , let $\kappa_m(\pi)$ retain the entries $1, \dots, m$ and replace every larger entry by a common symbol L . We call the resulting word a *skeleton*.

Define a map F_m on skeletons by the following symbolic stack procedure. An input L is always pushed. For an input $x \in [m]$, pop while at least two stack tokens belong to

$$\{x+1, x+2, \dots, m, L\},$$

then push x . Flush at the end.

Lemma 3.1 (large-label factorization). *For every permutation π ,*

$$\kappa_m(s_{123,132}(\pi)) = F_m(\kappa_m(\pi)).$$

Consequently, $\kappa_m(s_{123,132}^r(\pi)) = F_m^r(\kappa_m(\pi))$ for all $r \geq 0$.

Proof. Suppose an input value $y > m$ forces a pop. The popped stack top must also be larger than m . Otherwise the top is some $z \leq m < y$, while illegality requires two entries below z that exceed y . Those three current stack entries already form 123 or 132, a contradiction.

After all values greater than m are identified with L , a pop caused by a large input can be delayed. Such a pop only moves a top L to the output

and cannot cross a small stack entry. Passing later large inputs changes only the order of indistinguishable L tokens. If a small input arrives, every delayed L exceeds it. The actual execution retains at least the most recently pushed large entry, so the delayed extras are popped before a small token can cross them. If no small input remains, the final flush does the same.

Delay every large-caused pop in this manner. Every large input is then an immediate symbolic push. For a small input x , Lemma 2.1 depends only on the number of tokens in $\{x + 1, \dots, m, L\}$, giving exactly the rule defining F_m . This proves the one-step identity, and induction gives the iterated identity. $\square$

Corollary 3.2. *For $m = \lfloor (N - 1)/2 \rfloor$, membership in M_N depends only on the skeleton. If Q_N is the set of qualifying skeletons, then*

$$|M_N| = |Q_N|(N - m)!.$$

Proof. Only the labels $1, \dots, m$ occur in the half-decreasing test. Thus Lemma 3.1 and Berlow's theorem show that the first periodic iterate depends only on the skeleton. Each skeleton has exactly $(N - m)!$ lifts, one for each assignment of the large labels. $\square$

4. EXACT ENUMERATION AND CERTIFICATE

A skeleton is specified by choosing the positions of the m small labels and permuting those labels. Hence there are exactly

$$\binom{N}{m} m! = \frac{N!}{(N - m)!}$$

skeletons. The enumeration applies F_m until the first half-decreasing iterate and accepts exactly those attaining Zhang's bound $2m$.

Table 1 reports the complete reconstruction through length 14. In particular, it independently reproduces all relations in the range $n \leq 6$ reported by Zhang.

Two discovery programs implement different enumeration and evaluation routes. Enumerator A uses lexicographic multiset permutations, increasing large-label lifts, the two-larger-elements rule, and the half-decreasing stopping test. Enumerator B uses recursive skeleton generation, decreasing large-label lifts, an explicit pair search for a forbidden push, and literal cycle detection. They agree at every length in the table.

A third Python verifier imports neither discovery program nor their outputs. It parses the serialized JSON certificate, regenerates every skeleton, applies the symbolic map, checks all table entries, and rejects malformed or incomplete input. In the decisive range it checks

$$(2) \quad Q_{14} = \{wL : w \in Q_{13}\}.$$

TABLE 1. Certified skeleton and permutation counts.

N	m	total skeletons	$ Q_N $	$ M_N $
1	0	1	1	1
2	0	1	1	2
3	1	3	1	2
4	1	4	1	6
5	2	20	1	6
6	2	30	1	24
7	3	210	5	120
8	3	336	5	600
9	4	3,024	44	5,280
10	4	5,040	44	31,680
11	5	55,440	561	403,920
12	5	95,040	561	2,827,440
13	6	1,235,520	9,378	47,265,120
14	6	2,162,160	9,378	378,120,960

Proof of Theorem 1.1. At both lengths, $m = 6$. Complete enumeration gives $|Q_{13}| = |Q_{14}| = 9,378$. Corollary 3.2 yields

$$|M_{13}| = 9,378 \cdot 7! = 47,265,120$$

and

$$|M_{14}| = 9,378 \cdot 8! = 378,120,960.$$

The second integer is eight times the first. $\square$

5. THE EIGHT-TO-ONE MAP

For $q \in S_{13}$ and $j \in [14]$, define

$$E_j(q) = (q_1 + \mathbf{1}_{q_1 \geq j}) \cdots (q_{13} + \mathbf{1}_{q_{13} \geq j})j.$$

Proposition 5.1. *The map*

$$D : M_{14} \longrightarrow M_{13}, \quad D(\pi) = \text{std}(\pi_1 \cdots \pi_{13}),$$

is eight-to-one. Its fiber over $q \in M_{13}$ is

$$\{E_j(q) : j = 7, 8, \dots, 14\}.$$

Proof. Equation (2) says that every qualifying length-14 skeleton ends in L , and deleting that token gives a qualifying length-13 skeleton. Therefore the last value of $\pi \in M_{14}$ lies in $\{7, \dots, 14\}$, and deletion followed by standardization gives an element of M_{13} . Conversely, each $E_j(q)$ with $j \in \{7, \dots, 14\}$ has skeleton $\kappa_6(q)L$ and belongs to M_{14} . These eight permutations are distinct, satisfy $D(E_j(q)) = q$, and exhaust the fiber because the last entry of every preimage must be large. $\square$

6. LIMITATIONS AND REPRODUCIBILITY

The certified set identity (2) concerns only lengths 13 and 14. Its apparent analogue at all lengths would prove (1), but no all- n proof is claimed here.

The certificate and verifier use only exact integers and complete finite enumeration. No floating-point value, random search, modular comparison, or heuristic cutoff enters the result. The full verification command is

```
code/verify_certificate.py certificates/n1_to_n7_counts.json
```

The computational discovery and manuscript preparation used an AI coding assistant. All decisive claims were converted to a serialized exact certificate, independently reconstructed by separate verifier code, and subjected to malformed-input tests. No proof assistant such as Lean, Coq, or Isabelle was used.

REFERENCES

1. Katalin Berlow, *Restricted stacks as functions*, Discrete Mathematics **344** (2021), no. 11, 112571.
2. Owen Zhang, *The order of the (123, 132)-avoiding stack sort*, Enumerative Combinatorics and Applications **5** (2025), no. 3, Article S2R18.

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